

DevOps2 Endterm

Single-node Kubernetes platform

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Topology

VM 192.168.10.12 · Ubuntu 25.10 · arm64 · 4 CPU

Kubernetes 1.32 · kubespray · containerd ·
Calico CNI · CoreDNS · iptables kube-proxy



SSH port-forward

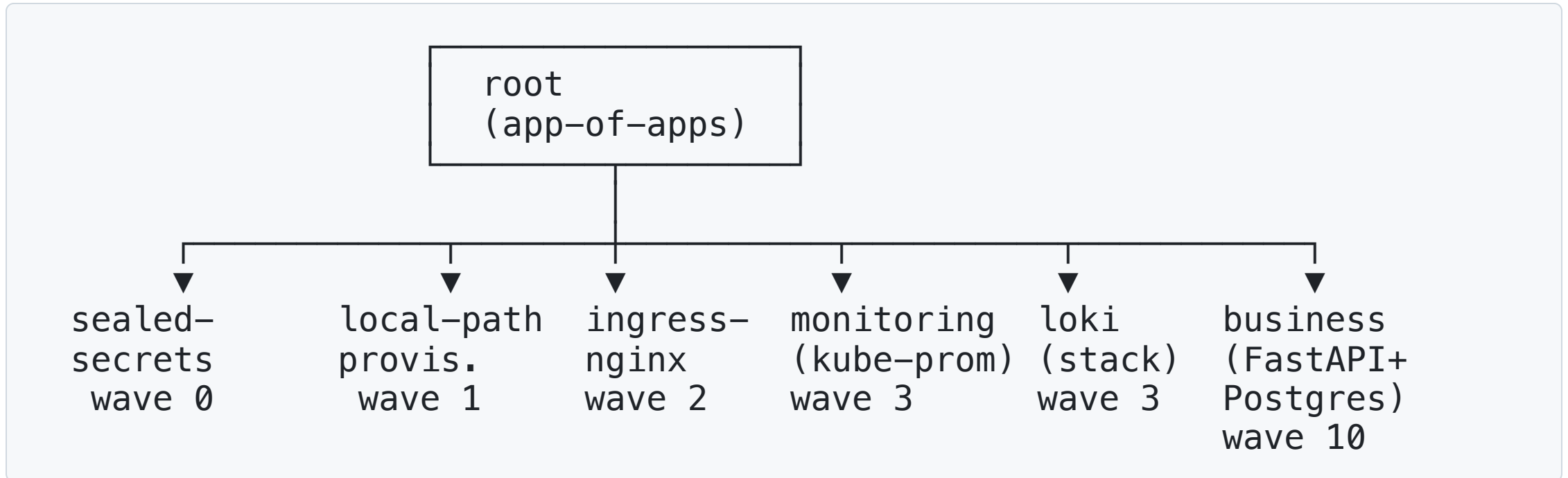
Mac (kubectl + git)

1 node = control-plane + worker · taint removed · single-node mode

Components by layer

Layer	Component	Purpose
Cluster	kubespray + Ansible	declarative cluster install
Runtime	containerd, Calico, CoreDNS	CRI / CNI / DNS
GitOps	ArgoCD v3.0.0	app-of-apps from <code>mkaIybek/devops2-endterm</code>
Secrets	sealed-secrets	encrypt creds at rest in git
Storage	local-path-provisioner	dynamic PVCs (default SC)
Ingress	ingress-nginx (NodePort 30080)	north-south traffic
Metrics	Prometheus + Alertmanager + Grafana	kube-prometheus-stack

ArgoCD app-of-apps



Sync-waves enforce ordering · `automated{prune,selfHeal}` · `ServerSideApply`

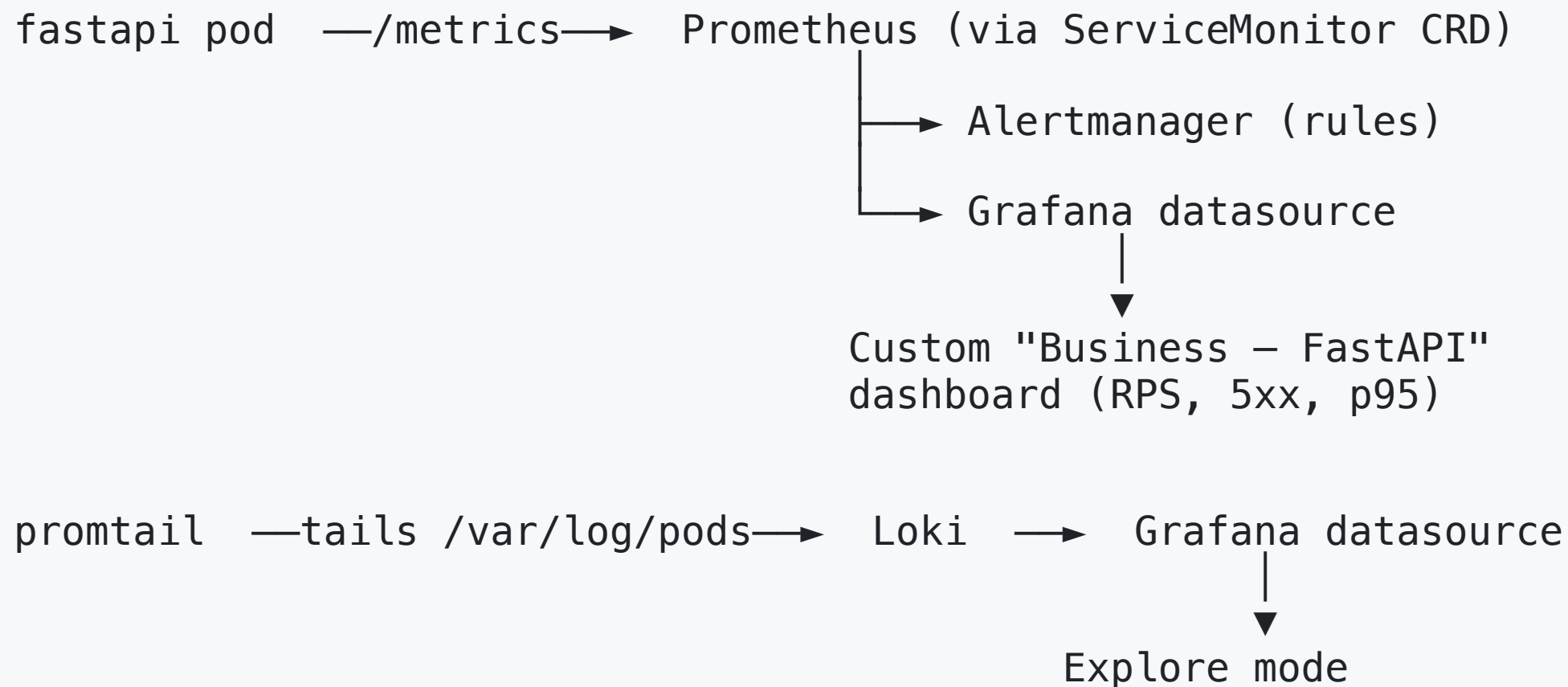
Security guardrails (Q5)

Control	Where
NetworkPolicy default-deny + 6 explicit allows	business ns
SecurityContext runAsNonRoot, readOnlyRoot, drop ALL caps	every container
ResourceQuota 2 cpu / 2 Gi req · 4 cpu / 4 Gi limit	business ns
LimitRange per-container defaults	business ns
PodDisruptionBudget minAvailable: 1	fastapi
SealedSecret encrypted DB creds in git	bitnami sealed-secrets
RBAC kubeconfig with X.509 client cert	kubespray PKI

Business app — request path



Observability — metrics flow



`prometheus-fastapi-instrumentator` → RED metrics for free

Live access

```
export KUBECONFIG=$(pwd)/kubeconfig

# ArgoCD UI          admin / v0Hjzf9EQ8lHFaSa
kubectl -n argocd port-forward svc/argocd-server 8081:443

# Grafana            admin / admin
kubectl -n monitoring port-forward svc/monitoring-grafana 3000:80

# Business app (чеpez VM)
ssh root@192.168.10.12 'curl -s -H "Host: business.local" http://127.0.0.1:30080/items'
```


Demo scripts

```
bash scripts/verify.sh           # full health snapshot
bash scripts/demo-netpol.sh      # prove NetworkPolicies deny
bash scripts/demo-zero-downtime.sh 2.0.0 # rolling update with 0 downtime
```

demo-zero-downtime.sh запускает curl loop в фоне · бампит image tag в git · ArgoCD reconcile → RollingUpdate maxUnavailable=0 → 100% 200 OK во время rollout

Defense Q-map

#	Q	File
1	Topology	docs/01-topology.md
2	Deployment process	docs/02-deployment.md
3	Access / config mgmt	docs/03-access.md
4	Business architecture	docs/04-architecture.md
5	Workload security	docs/05-security.md
6	Networking	docs/06-networking.md
7	Storage	docs/07-storage.md
8	Rollout	docs/08-rollout.md
9	Custom extensions	docs/09-extensions.md